

PH 508: Magnetism and Magnetic Materials

Course Code	PH 508
Course Name	Magnetism and Magnetic Materials
L-T-P-C	3-0-0-3
Prerequisites	PH 301 Quantum Mechanics
Intended for	UG/PG Distribution: Elective
Approval	8th Senate (Oct 2015)

Course Contents

- **Introduction**
- History of magnetism, Magnetic units, Classical and quantum mechanical model of magnetic moment of electrons, magnetic properties of free atoms. [3 Lectures]
- **Types of magnetism**
- Classification of magnetic materials, Theories of Diamagnetism, Paramagnetism, Theories of ordered magnetism, Quantum theory of magnetism: electron-electron interactions, localized electron theory, itinerant electron theory. [8 lectures]
- **Magnetic interactions**
- Origin of crystal field, Jahn Teller effect, Magnetic dipolar interaction, Origin of exchange interaction, Direct exchange interactions, Indirect exchange interactions in ionic solid and metals, double and anisotropic exchange interaction. [5 lectures]
- **Magnetic domains**
- Development of domain theory, Bloch and Neel Wall, Domain wall pinning, Magnons, Bloch's law, Magnetic anisotropy, magnetorestriction. [5 Lectures]
- **Competing interactions and low dimensionality**
- Frustration, Spin glass, superparamagnetism, one and two dimensional magnets, Thin film and multilayers, Heisenberg and Ising models. [4 lectures]
- **Novel magnetic materials**
- Colossal and giant magnetoresistive materials, magnetic refrigerant materials, Shape memory alloys, multiferroics, spintronics devices and their application in magnetic storage. [7 lectures]
- **Measurements techniques**
- Production and measurement of field, magnetic shielding, Faraday balance, AC susceptometer, Vibration sample magnetometer, torque magnetometer, SQUID magnetometer, Experimental method in low temperature. [8 lectures]

Text Books

1. B. D. Cullity and C. D. Graham, **Introduction to magnetic materials**, John Wiley & Sons, 2011.
2. D. Jiles, **Introduction to magnetism and magnetic materials**, Taylor and Francis, CRC Press, 1998.

References

1. K. H. J. Buschow and F. R. de Boer, **Physics of Magnetism and Magnetic Materials**, Kluwer Academic Publishers, 2003.
2. Stephen Blundell, **Magnetism in Condensed Matter**, Oxford University Press, 2001.
3. Mathias Getzlaff, **Fundamentals of Magnetism**, Springer, 2008.